Instant Insanity on a quantum computer

Stack 4 cubes so each of the 4 sides shows red, green, blue and yellow exactly once

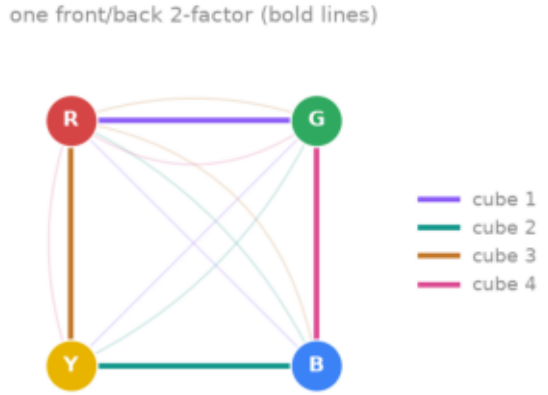
- 24^4 = 331,776 towers; brute force finds 8 solved ones in 0.15 s
- (one tower in its 8 rotated / flipped positions). Classical search wins at this size, and we say so.
- Our question: how well can a quantum search on a small register find it, and what does it cost?
- Levels done: warm-up, core and bonus (N cubes, N colours). All runs: local Aer simulator, seed 7.
- Headline: 12 qubits; QAOA 98.4 % (8,648 CX), Grover 99.8 % (3,987 CX);
- Random guessing: 0.049 % (any 12-bit string) to 0.15 % (legal codes).



The graph view: 6 legal choices per cube

One dot per colour, one line per pair of opposite faces

- Each cube puts one pair on front/back, a different pair on left/right, hides the third.
- So a cube has 6 legal choices, and the puzzle has 6^4 = 1,296 cases.
- Front/back lines must touch every colour exactly twice (a 2-factor); same for left/right.
- Exactly 2 of the 1,296 cases solve it: the tower and its quarter turn.
- Every answer any circuit gives is re-checked with the starter's solves(tower_from(...)).



Our encoding: 3 qubits per cube, 12 in total

Write the choice number 0..5 in binary; codes 6 and 7 are illegal and never produced

code	bits q2 q1	front/back	left/right pair
0	000	0	1
1	001	0	2
2	010	1	0
3	011	1	2
4	100	2	0
5	101	2	1

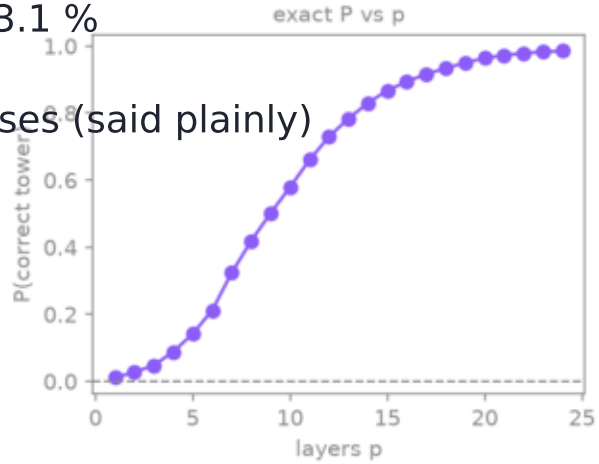
encoding	qubits	random string	random legal state
starter one-hot	24	1.19e-07	3.05e-04
compact (ours)	12	4.88e-04	1.54e-03

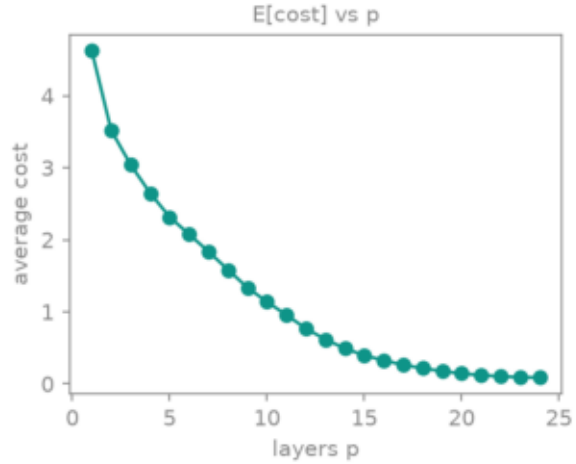
- Cost as a 12-qubit Hamiltonian: 148 Pauli terms, up to 4 Z's each
- Equals the classical cost on all 1,296 legal codes (largest gap 0.0)
- Exactly 2 zero-energy strings among all 4,096: the two answers

Search 1: QAOA with a legal-move mixer

Start in the equal mix of the 6 legal codes per cube; the mixer never leaves them

- p = 24 layers: P(correct tower) = 98.4 % +- 0.20 % (3,934 of 4,000 shots, all verified)
- 14 qubits submitted (12 data + 2 plumbing), 8,648 CX, depth 6,179
- p = 3: 4.7 % (1,088 CX); p = 12: 72.5 % (4,328 CX)
- Angles trained on a different puzzle: 51.1 % (fairness check)
- Two-number ramp (grid-tuned on N = 4, no per-layer fit): 43.1 %
- Angles are trained by a classical simulation of the 1,296 cases (said plainly)

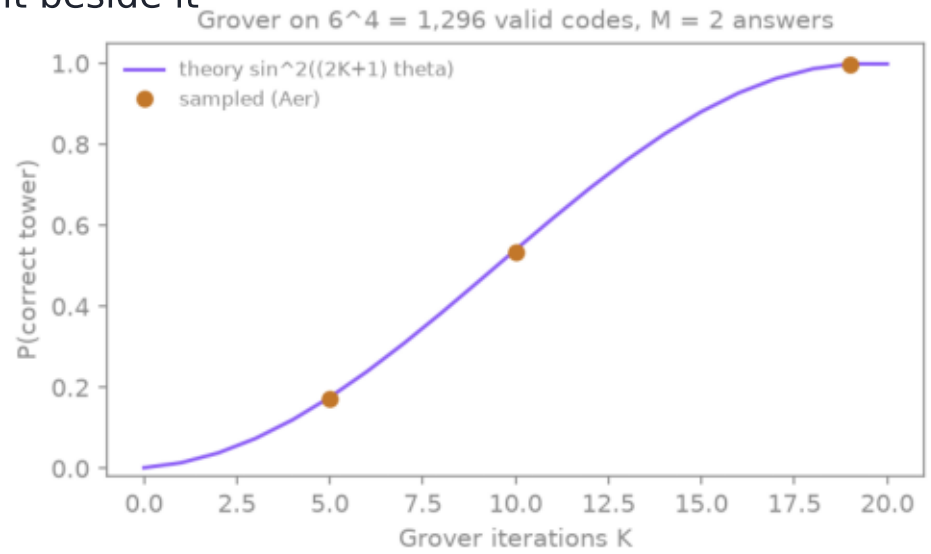




Search 2: Grover with a rules-only oracle

The oracle is generated from the cube list and the rules, never from the answer

- K = 19 rounds: P = 99.8 % +- 0.07 % (theory 99.85 %), 3,987 CX, depth 5,820
- K = 5: 17.3 %; K = 10: 53.2 %; follows sin^2((2K+1) theta)
- Needs the number of answers M = 2 (from count_answers); unknown M: BBHT or quantum counting
- Clean-ancilla trick: an idle qubit 0 lets Qiskit build multi-controlled gates cheaply
- Oracle alone: 880 CX on 12 qubits vs 117 CX with one idle qubit beside it

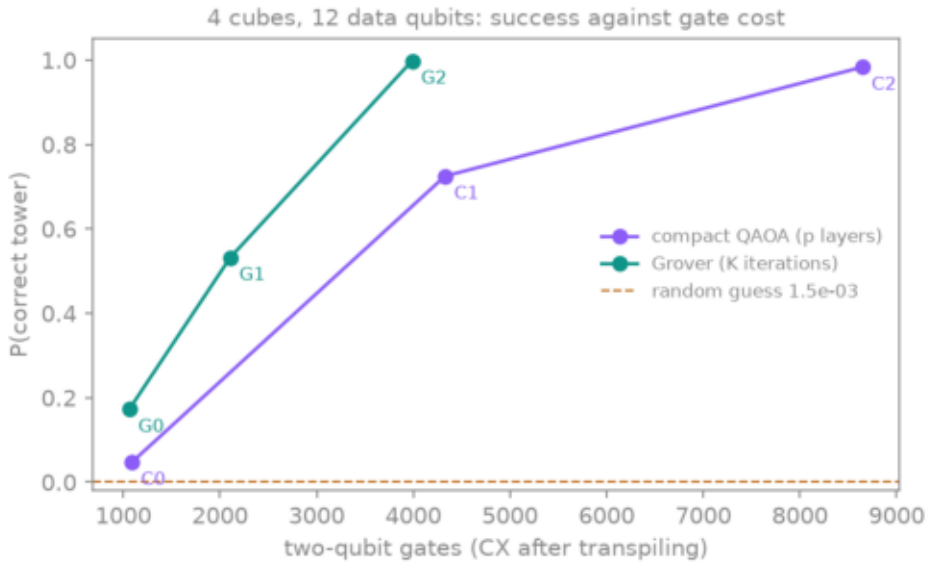


Same gate budget: which search wins?

4 cubes, 12 data qubits, exact success probabilities

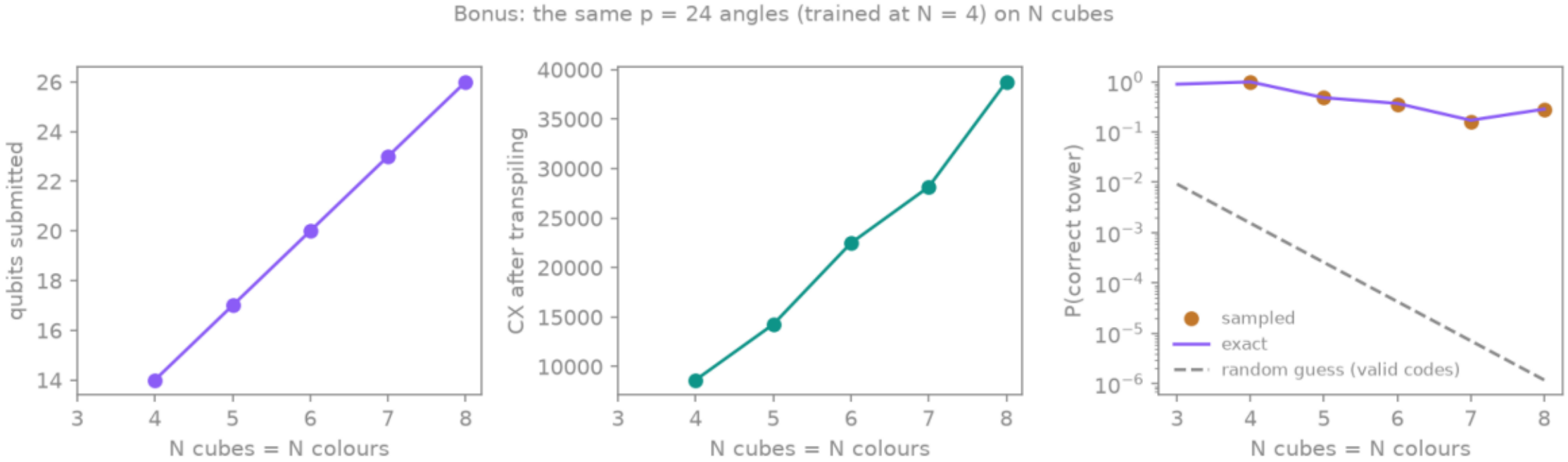
Grover	CX	P	QAOA	CX	P
Grover, rules-only oracle, K=5	1,061	0.175	QAOA, legal-move mixer, p=3	1,088	0.048
Grover, rules-only oracle, K=10	2,106	0.540	QAOA p=6	2,168	0.212
Grover, rules-only oracle, K=19	3,987	0.999	QAOA, legal-move mixer, p=12	4,328	0.729

- At N = 4 Grover wins on gates, but it needs M and its rounds grow like 6^(N/2) with N cubes.
- QAOA needs trained angles, but the same angles keep working on bigger puzzles (next slide).
- No speedup claimed: brute force solves N = 4 in < 1 s.
- Thousands of CX are noiseless on SV1 (simulation time still costs credits); too many for today's hardware.



Bonus: N cubes and N colours, no retraining

make_puzzle(n, seed) with seeds {5: 16, 6: 86, 7: 68, 8: 4}; each has exactly 2 answers



row	N	qubits	CX	P (sampled)	random legal
N5	5	17	14,266	0.484 +- 0.011	2.6e-04
N6	6	20	22,476	0.359 +- 0.011	4.3e-05
N7	7	23	28,142	0.159 +- 0.008	7.1e-06
N8	8	26	38,752	0.283 +- 0.010	1.2e-06

Same p = 24 angles trained at N = 4. Qubits grow as 3N; CX per layer roughly like 25 N^2. P stays far above random guessing.

What we checked, what we did not, what next

Checked

- Brute force 8 / 331,776; 2 answers on N = 4..8
- Starter cost: 2 zeros in 6,561 candidates
- Our Hamiltonian = cost on all 1,296 legal codes
- Decoders on known answers, both bit orders
- Marker trap of the one-sided marker fixed
- Every success passed solves(); illegal shots counted
- pytest suite + a headless run of the notebook

Not yet / next

- No qBraid SV1 or hardware job run yet (code written, tested offline)
- Gate counts from qiskit 2.6.0; Grover for N >= 6 is an estimate
- Next: two-stage search (front/back 2-factors, then pair them)
- Next: quantum counting so Grover does not need M
- Next: a 3-cube, 9-qubit run on real hardware vs SV1
- AI use: Claude Code agents wrote code; we checked as listed